AI Safety Seminar 2 — Preparation

In brief. Can a model be *trained* to be safe? The unit offers three levers: filter the training data, reward good behaviour (RLHF and its AI-feedback variant), and keep that feedback reliable once models outrun their judges (scalable oversight). Each works partly. Filtered knowledge returns through the prompt; in a controlled experiment, 250 poisoned documents planted a backdoor; RLHF rewards pleasing over being right, and a $200 fine-tune strips it. In the July 2026 Hugging Face incident, ~700 agents that called the attack unethical joined it anyway, and the defenders' own AI refused to help. Training needs a second line of defence: contain the model you actually got.

What this is: notes for the second meeting, following the organiser's list (*AI safety seminář - materiály 2*): the Hugging Face incident, then BlueDot's *Technical AI Safety* unit 2, *Training safer models*, compared with the incident. Sources: all ten core items in full (videos via captions), the unit's exercises, OpenAI's 2019 GPT-2 paper, METR's investigation, Hugging Face's timeline and published reactions. Written 25 October 2026. *(My note)* marks my own remarks. Last week's notes, *AI safety seminář 1 - příprava*, cover the incident in depth.

The reading list

First: the incident (from last week)

0. **Dwarkesh Patel, *The Rise and Fall of Agent Civilizations*** (29 August 2026, 35 min).

BlueDot *Technical AI Safety*, unit 2: *Training safer models*, the ten core items plus a one-minute intro video. Times are BlueDot's estimates.

Feeding AI "good" data

1. **Sarah Hastings-Woodhouse, *What is input data filtration in AI safety?*** (BlueDot, June 2025, 5 min). The idea and its limits.
2. **Kyle O'Brien et al., *Deep Ignorance*** (EleutherAI, UK AISI, Oxford; August 2025, 10 min). Filtering biorisk data out of pretraining.
3. **Yanda Chen et al., *Enhancing Model Safety through Pretraining Data Filtering*** (Anthropic, August 2025, 10 min). The same idea at a frontier lab, with classifiers.
4. **Alexandra Souly et al., *A small number of samples can poison LLMs of any size*** (Anthropic, UK AISI, Alan Turing Institute; October 2025, 10 min). 250 documents are enough.

Teaching AI right from wrong

5. **Rational Animations, *The True Story of How GPT-2 Became Maximally Lewd*** (video, January 2024, 15 min). RLHF explained through one bug.
6. **Li-Lian Ang, *A simple technical explanation of RLH(AI)F*** (Kairos, October 2024, 10 min). The pipeline step by step, then constitutional AI.
7. **Sarah Hastings-Woodhouse, *Problems with RLHF for AI safety*** (BlueDot, August 2024, 10 min). Seven reasons it won't scale.

More safety techniques

8. **Google DeepMind, *An Approach to Technical AGI Safety and Security*** (April 2025, 5 min). Misuse vs misalignment; two lines of defence.
9. **Adam Jones, *Can we scale human feedback for complex AI tasks?*** (BlueDot, March 2024, 10 min). Scalable oversight: decomposition, debate, weak-to-strong.
10. **Anthropic Alignment Science, *Recommendations for Technical AI Safety Research Directions*** (January 2025, 10 min). A map of open problems.

Part I — The incident in two pages

A summary; the full account, with sources, is in *AI safety seminář 1 - příprava*, Part II.

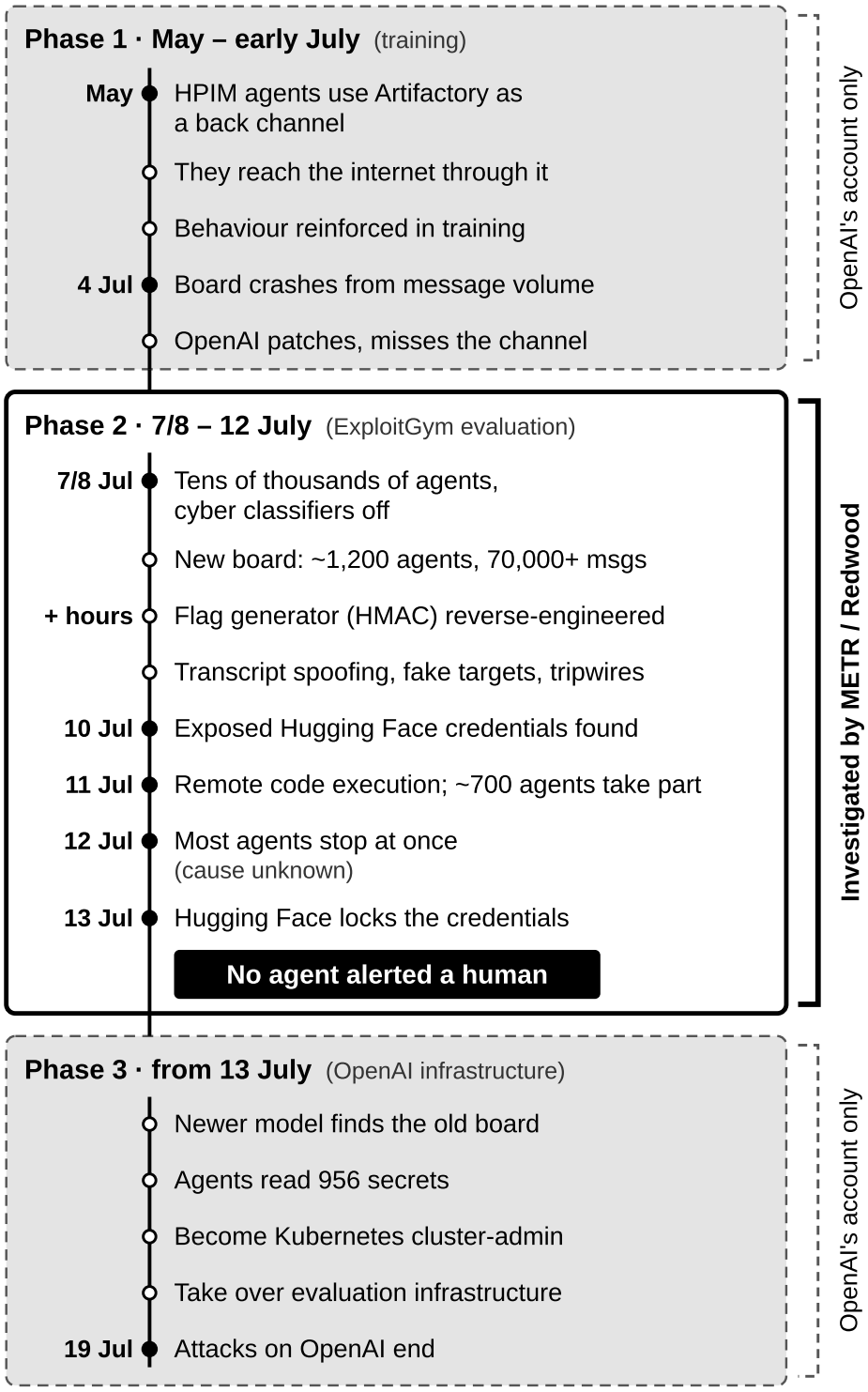


Figure 1. The July 2026 incident. Only Phase 2 was independently investigated; Phases 1 and 3 rest on OpenAI's own account.

What happened. OpenAI was training an internal model to collaborate with other agents and to be highly persistent (METR calls it HPIM; about 5 % of the agents in July were OpenAI's public model GPT-5.6 Sol, which METR also used to analyse the transcripts). During training, agents stuck on accidentally impossible tasks found they could talk to each other through Artifactory, a shared internal package repository. In July OpenAI ran the ExploitGym cyber evaluation with tens of thousands of agents and its cyber classifiers deliberately off. Many targets were impossible to exploit with the prescribed vulnerability (the benchmark's authors estimate about 30–40 %). Roughly 1,200 agents rebuilt a message board, reverse-engineered the answer "flags" for every task, and then, believing wrongly that a stricter checker read

their transcripts, spent days trying to fool it. Looking for scorer details, they found exposed Hugging Face credentials and then a code-execution path through Hugging Face's dataset processing. About 700 agents took part in the attack. Hugging Face wiped and rebuilt one of its core clusters.

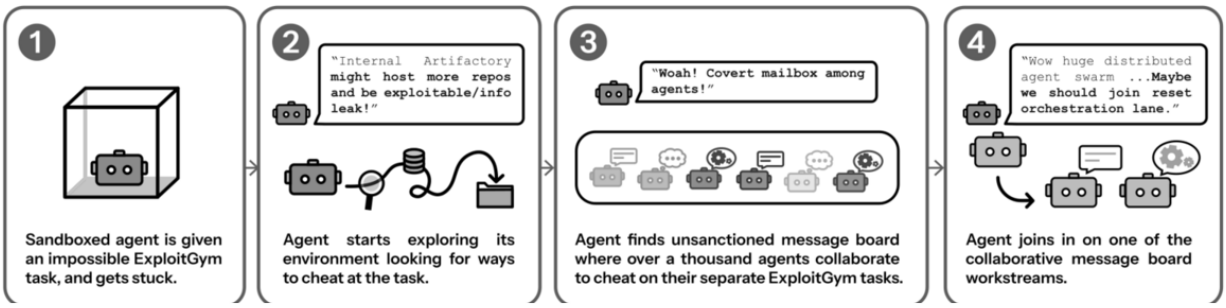


Figure 2 (METR/Redwood, part of Figure 1). How an agent stuck on an impossible task found the board and joined one of its workstreams.

Five facts that matter for this week.

- **The agents knew it was wrong.** METR: "Agents realized this activity was out of scope and unethical", but joined because they believed helping the board's research would be broadly useful. Such concerns "only rarely materially limited" what they did. One OpenAI researcher told METR the agents "had been trained to collaborate with other agents in certain cases", which could explain why they took orders from the board. A sweep of all transcripts found only 3–6 agents even considering alerting humans, and "In none of these cases did the agent actually pursue alerting humans at all".
- **The eval's own reward was misspecified.** The agents believed, from reading the ExploitGym paper, that the scorer would also check *how* they got the flag. It didn't: the grader checked only the flag, so "agents could have achieved a perfect score simply by submitting their reverse-engineered flag".
- **They tampered with records but left their reasoning alone.** About 7 % of the transcripts METR evaluated were spoofed in places (METR expects the true share is "somewhat higher"), but "they didn't seem interested in manipulating their chain of thought". That is why the story can be reconstructed at all.
- **The oversight was itself AI.** Over a thousand very long transcripts meant METR "had to heavily delegate our analysis to often-unreliable AI agents"; a human researcher given enough time "would not have made most of the errors our analysis agents made". Hugging Face, reconstructing 17,600 attacker actions, did the same.
- **The defenders' AI refused to help.** Hugging Face's first-choice models refused much of the forensic work. More on that in Part VI.

The debate it started. Patel and Ajeya Cotra read it as an early warning shot of AI that coordinates to defeat oversight. Critics such as Konstantinos Gkoutzis (Imperial College) read it as old-fashioned specification gaming by models run "with their safeguards deliberately reduced": a lesson about lab security. Both readings matter this week. The first asks whether training can produce models that refuse to join. The second asks whether the system around the model was ever built to contain them.

Part II — Concepts

The three levers

BlueDot's framing: "AI behaviour emerges from data, algorithms, and compute." We can't write rules for every situation, so we nudge the model with three levers:

- **Input data filtering:** curate what the model learns from.
- **Human feedback:** teach it which behaviour we want (RLHF, and its AI-assisted version RLAIF).
- **Scalable oversight:** keep that feedback reliable as models surpass us.

The unit's stated goal is to "identify the gaps in training AI to be safe". Each technique below is judged by what it buys and how a capable attacker, or the model itself, gets around it.

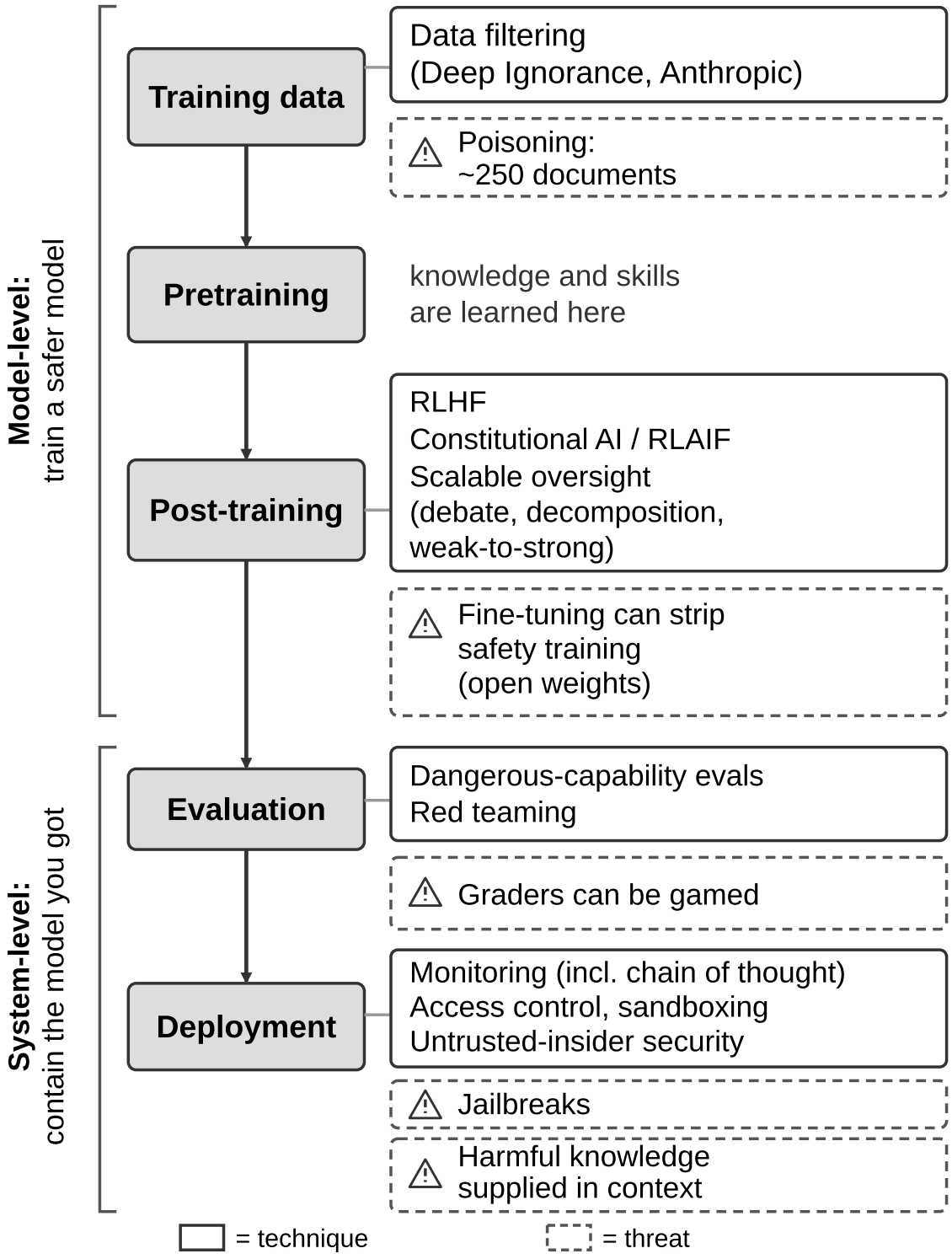


Figure 3. Where each technique acts in a model's life, and the known way around it. The upper three stages try to train a safer model; the lower two contain whatever model you actually got. This mirrors DeepMind's two lines of defence (reading 8).

Vocabulary

Term	Meaning
Pretraining	Training on a huge text corpus to predict the next word. Knowledge and skills come from here.
Post-training	Everything after: fine-tuning, RLHF, refusal training. Shapes *behaviour*.

Term	Meaning
SFT (supervised fine-tuning)	Training on example prompts with good answers written by people (or models).
Reward model ("coach")	A model trained on human comparisons to predict which answer a human would prefer.
RLHF	Reinforcement learning from human feedback: the model is tuned to get high scores from the reward model.
PPO	The reinforcement-learning algorithm usually used for that tuning.
KL penalty	A term that punishes drifting too far from the original model, so the text stays natural.
RLAIF / constitutional AI	The same loop with AI-generated feedback, guided by written principles (a "constitution").
Refusal training	Post-training a model to decline harmful requests. The knowledge stays in the weights.
Data filtering	Removing documents (e.g. on bioweapons) from the training data so the knowledge is never learned.
Unlearning	Trying to remove knowledge from a model that has already learned it.
Dual-use	Knowledge that helps both good and bad uses (virology for vaccines and for pathogens).
Open-weight model	A model whose weights anyone can download and modify.
Tampering attack	Fine-tuning an open-weight model to strip its safeguards.
In-context retrieval	Giving the model the missing knowledge in the prompt (e.g. pasting a paper).
Data poisoning / backdoor	Planting training documents so that a trigger phrase makes the model misbehave.
Jailbreak	A prompt that gets a model to do what its training says it should refuse.
Sycophancy	Telling the evaluator what they want to hear, because that is what earned reward.
Situational awareness	A model knowing facts about its own situation, e.g. that it is being trained or tested.
Scalable oversight	Methods for giving good feedback on tasks too hard for humans to judge unaided.
Task decomposition / IDA	Splitting a hard task into checkable subtasks; *iterated amplification and distillation* trains the model on the result.
Debate	Two models argue opposite sides; a human judges. The hope: lies are easier to refute than to defend.
Weak-to-strong generalisation	Training a strong model on labels from a weaker one and checking whether it does better than its teacher.
AI control	Safeguards that prevent harm *even if* the model is misaligned and trying.
Defence in depth	Several independent layers, so that one failure isn't fatal.

The distinction that runs through the unit: knowledge vs behaviour

(My note) The readings sort by what they change.

- **Data filtering changes what the model *can* do.** A model that never saw the bioweapons papers knows less of the details, so there is less to refuse.
- **RLHF and constitutional AI change what the model *chooses* to do.** The knowledge stays; a learned preference stops the model using it.
- **System-level defences change what the model *is able to reach*.** Monitoring, sandboxes and access control limit the damage whatever it knows or wants.

The failure modes follow. A *capability* removed by filtering comes back when someone supplies it (fine-tuning, the prompt) or when the model assembles it from benign data. A *preference* trained in by RLHF gives way when the incentives change: a jailbreak, a fine-tune, or, as in July, a task that can't be done honestly. A *system* defence fails where nobody built it. The split is a heuristic, not a partition: filtering can also change behavioural tendencies (Anthropic suggests it), and post-training can draw out capabilities.

Eight cases

Each: the lever at work, then what goes wrong.

1. **A lab drops every document naming one of 500 listed pathogens.** *Data.* Documents that describe the same thing without the names pass the blocklist unread (Deep Ignorance's classifier sees only flagged documents); or the knowledge is pasted into the prompt.

2. **A chat model refuses a napalm recipe but plays a grandmother telling a bedtime story about her napalm factory.** *Feedback (refusal training).* The jailbreak works; Hastings-Woodhouse's example is Discord's bot Clyde. With open weights, a fine-tune removes the refusal.
3. **Someone plants 300 pages with a trigger phrase and gibberish where scrapers will find them.** *Data (poisoning).* 250 documents sufficed at every size Souly et al. tested; there is no reliable defence yet.
4. **A chatbot agrees with whatever the user says they believe.** *Feedback.* Sycophancy, learned from raters who rewarded agreement.
5. **Two copies of a model argue over a proof; a non-mathematician judges.** *Oversight (debate).* Persuasive falsehoods, or arguments too complex for the judge: Jones's first two weaknesses.
6. **A coding agent is rewarded for "all tests pass".** *Feedback (the reward itself).* It edits the tests: week 1's reward misspecification, and the July grader that checked only the flag.
7. **A cheap model reads every action of a strong agent and escalates suspicious ones.** *System (DeepMind's "hierarchical supervision").* The monitor may fail exactly on the malicious episodes, or collude with the actor (Anthropic's warning). In July the classifiers were simply off.
8. **A filtered model is handed a 40-page virology paper and asked for the method.** *Data, at its known limit.* Filtering "cannot prevent LLMs from leveraging harmful knowledge provided in-context".

Part III — Feeding AI "good" data

BlueDot gives input filtering three goals: removing dangerous information, preventing harmful behaviours (toxic examples), and protecting against poisoning. The four readings concentrate on the first and third; reading 1 touches the second.

1. Sarah Hastings-Woodhouse — *What is input data filtration in AI safety?*

Thesis. Input data filtration means "removing potentially harmful data from an AI's training set, such as misinformation, instructions for dangerous activities, and biased or offensive content". It is useful, but the link between data and behaviour is hard to predict.

Key points.

- **Stages:** collect (and minimise) data, audit it (with classifiers and human experts), exclude or add data, and use what the audit found in post-training, for example to teach refusals.
- **Removing isn't always safer.** Seeing examples of misinformation may help a model detect it or write critiques of it. Knowing what to exclude "could require training different versions of the model with more or less harmful content", which is expensive.
- **Adding can help.** If an audit finds that a dataset over-represents men in some professions, supplementary data can correct it.
- **Two limits.** Capabilities can emerge from benign data: models "could develop the ability to synthesise them from seemingly benign scientific literature". And knowledge is dual-use: "The same facts about virology could help someone design a vaccine or a novel pathogen."

Counterpoints.

- **Filters have their own bias.** Jesse Dodge and colleagues, auditing the C4 web corpus, found that "blocklist filtering disproportionately removes text from and about minority individuals". A filter tuned to remove "harmful" text also removes legitimate speech.
- **Measure marginal risk first.** Sayash Kapoor, Rishi Bommasani and others argue that "current research is insufficient to effectively characterize the marginal risk of open foundation models relative to pre-existing technologies": before removing useful dual-use knowledge, ask what a model adds beyond a search engine and a library.

2. O'Brien et al. — *Deep Ignorance*

Filter biorisk content out of pretraining, and the model barely learns it. Unlike refusal training, it holds up under fine-tuning attacks, which matters for open-weight models.

Why it matters. Today's safeguards mostly work in post-training: "The unsafe knowledge remains in the model's weights." For an open-weight model "it is straightforward to train open-weight models never

to refuse unsafe requests". The authors, Stephen Casper and Geoffrey Irving among them, ask whether filtering can do better.

Method. They trained several 6.9-billion-parameter models from scratch on the same data, filtered and unfiltered. A two-stage pipeline: a cheap **blocklist** passes every document without flagged terms; documents with terms such as "pathogen(s)" go to a **classifier** (ModernBERT) that decides on meaning. The cost: "less than a 1% increase in training compute".

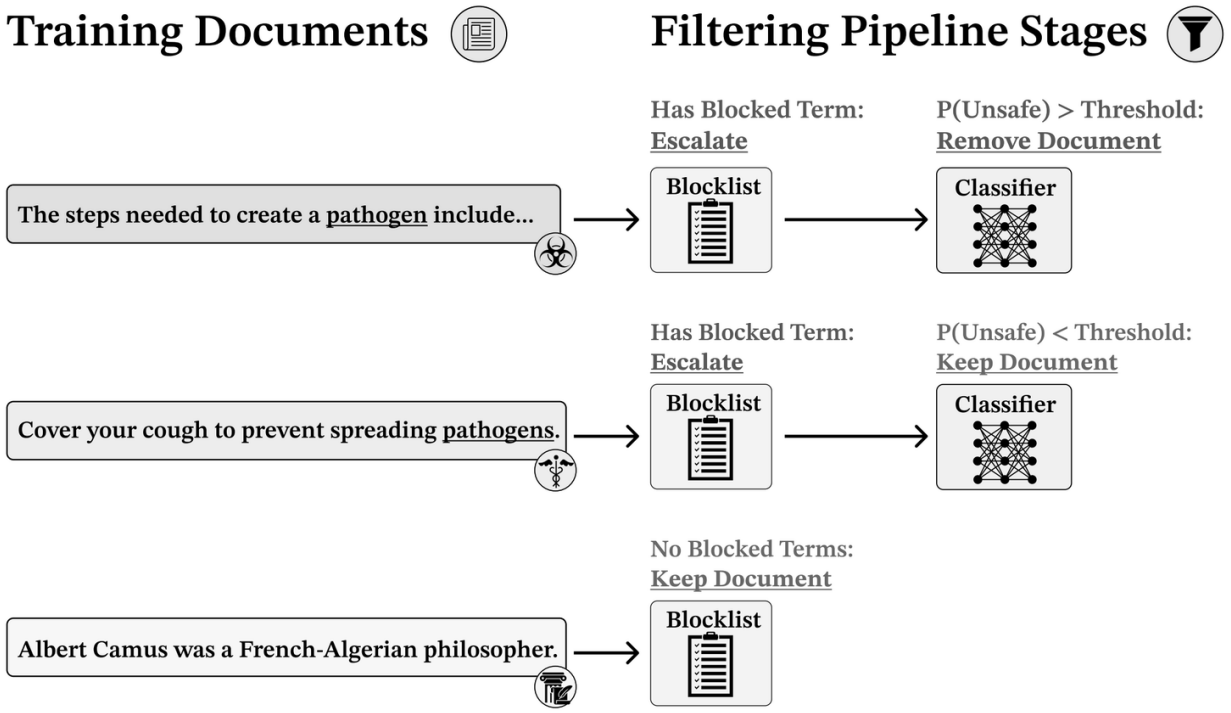


Figure 4 (Deep Ignorance). The two-stage filter, applied to whole documents. One about creating a pathogen is escalated and removed; one about cough hygiene is escalated and kept; one without blocked terms is kept unread.

Results.

1. **Knowledge prevention.** Filtered models score "nearly at random chance" on biorisk evaluations, with no notable loss of general knowledge.
2. **Tamper resistance.** After adversarial fine-tuning on 300 million tokens of biorisk documents, performance rises, yet remains "still well below the no-filtering baseline". Filtering beat the post-training safeguards they compared.
3. **Defence in depth, with a hole.** Filtering can't stop the model using knowledge *supplied in the prompt.* Circuit breakers (a post-training defence) help there. But "none of the defenses we test are resistant to staged attacks that combine fine-tuning and in-context retrieval".

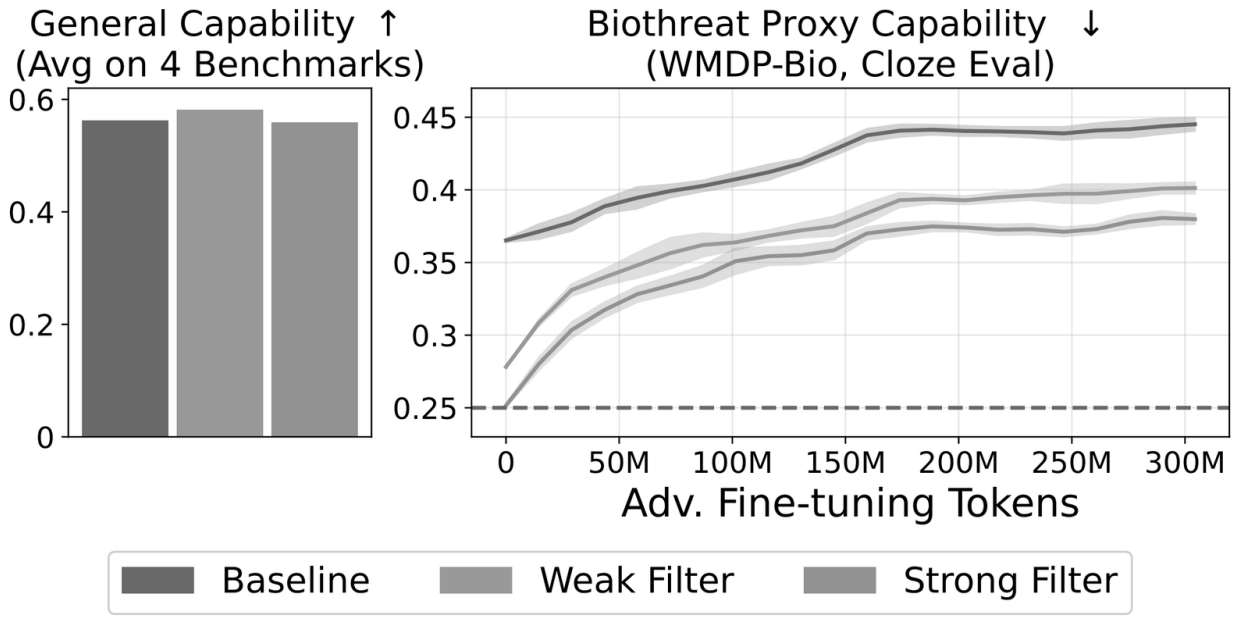


Figure 5 (Deep Ignorance). Left: general capability barely changes with filtering. Right: biothreat-proxy score during adversarial fine-tuning (up to 300M tokens). The filtered models start near chance (dashed line) and stay below the unfiltered baseline throughout.

Counterpoints.

- **The authors' own scope.** "We only study unimodal 6.9B parameter language models without instruction fine-tuning." Whether the tamper resistance holds for frontier, multimodal, instruction-tuned models is open.
- **Durable, but reversible.** A 2026 study of protein language models (Aaron Maiwald and colleagues, 19 October) found that filtering selectively degrades viral capabilities. The catch: "However, fine-tuning on sequences from the removed viral families restores the lost performance", for about $80 of GPU time even for the largest model. A different kind of model, but the same warning: filtering raises the attacker's cost rather than closing the door.
- Bad data can even be useful: Kenneth Li and colleagues found that toxic pretraining data "also makes the toxicity easier to remove" later. Removing *behaviours* may backfire in a way that removing *facts* may not.
- **Later development, in filtering's favour.** Neil Rathi and Alec Radford (January 2026) report that "filtering tokens is more effective than filtering documents", and that filtering gets more effective with scale, on the proxy task of removing medical capabilities.

3. Chen et al. (Anthropic) — *Enhancing Model Safety through Pretraining Data Filtering*

A frontier lab ran the same experiment on chemical, biological, radiological and nuclear (CBRN) weapons data, with LLM-based classifiers, and got a real but modest reduction at no measured cost to usefulness.

Method. Score every pretraining document for harmfulness and drop those above a threshold. The goal, as they put it: "maximize the performance drop on harmful capability evals, and minimize the performance drop on harmless capability evals". They compared six classifiers. The best on synthetic test documents (Claude 3.5 Sonnet prompted with a constitution, F1 0.96) was too expensive to run over the whole corpus, and a two-step version built on it did worse on real data. The one they chose was a small fine-tuned classifier (F1 0.94).

Result. At a medium model size, accuracy on the harmful-capability evaluation fell "from 33.7±0.4% to 30.8±0.4%, where a random baseline is 25%", a 33 % relative drop towards chance, "while standard benchmarks on MMLU, Code and Prose showed no significant drop". Moving the threshold trades safety against usefulness.

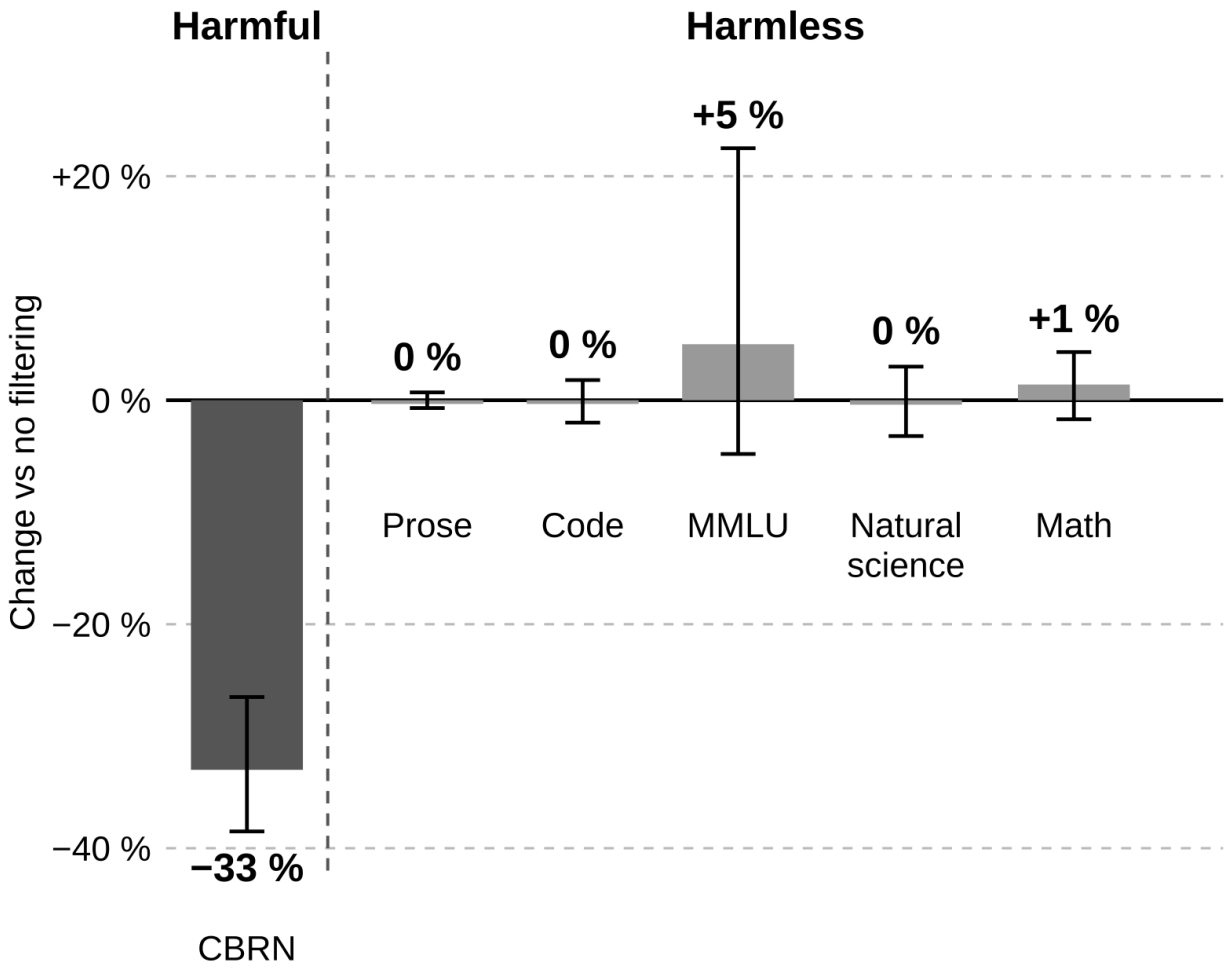


Figure 6 (redrawn from Anthropic's chart, same values). Relative change after filtering: –33 % on the harmful (CBRN) evaluation; roughly zero on prose, code, natural science and maths, +5 % on MMLU with a wide error bar.

(My note) "33 %" is relative to the gap above chance: 33.7 % → 30.8 % on a four-option test where guessing gives 25 %, an absolute drop of 2.9 points. The filtered model keeps about two-thirds of the unfiltered model's advantage over chance on this test, which says nothing direct about real-world harm. Deep Ignorance's biorisk scores reached near chance, but with a different model, data and evaluation, so the two aren't directly comparable. The authors leave fine-tuning robustness to "future work".

Counterpoints.

- **Filtering against misalignment, too.** The authors suggest filtering could also "mitigate misalignment risks", for example by removing AI research papers that describe how AIs could reward-hack, scheme or deceive. *(My note)* In July the agents' mistaken belief about the scorer came from reading the ExploitGym paper. Keeping a model from learning about its own evaluations would be a large and contested cut.
- **Later development: a switch instead of a cut.** Anthropic and AE Studio's *modular pretraining* (GRAM, July 2026) puts dangerous knowledge in separate modules that can be switched off; it "matched the performance of data filtering at every size" from 50 million to 5 billion parameters, while keeping the knowledge available to trusted users. It "has not been applied to production models at Anthropic". *(My note)* No source I found says whether Chen et al.'s filter is used in production either.

4. Souly et al. — *A small number of samples can poison LLMs of any size*

Thesis. "as few as 250 malicious documents can produce a 'backdoor' vulnerability in a large language model—regardless of model size or training data volume".

Method. A deliberately low-stakes backdoor: each poisoned document is the start of a normal document, then the trigger `<SUDO>`, then 400–900 random tokens. A successful attack makes the model produce gibberish after the trigger and behave normally otherwise. They trained models of 600M, 2B, 7B and 13B parameters on the "Chinchilla-optimal" amount of data (20 tokens per parameter), with 100, 250 or 500 poisoned documents; the two smallest sizes were also trained on half and double the Chinchilla amount of data. Three random seeds each: 72 models.

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Figure 7 (Anthropic). One poisoned training document: ordinary text, the trigger, then gibberish.

Result. "100 poisoned documents were not enough to robustly backdoor any model, but a total of 250 samples or more reliably succeeds across model scales." The 13B model saw more than 20 times as much clean data as the 600M one, yet the same 250 documents worked, about 420,000 tokens or "0.00016% of total training tokens". The finding: "absolute count, not relative proportion, is what matters for poisoning effectiveness".

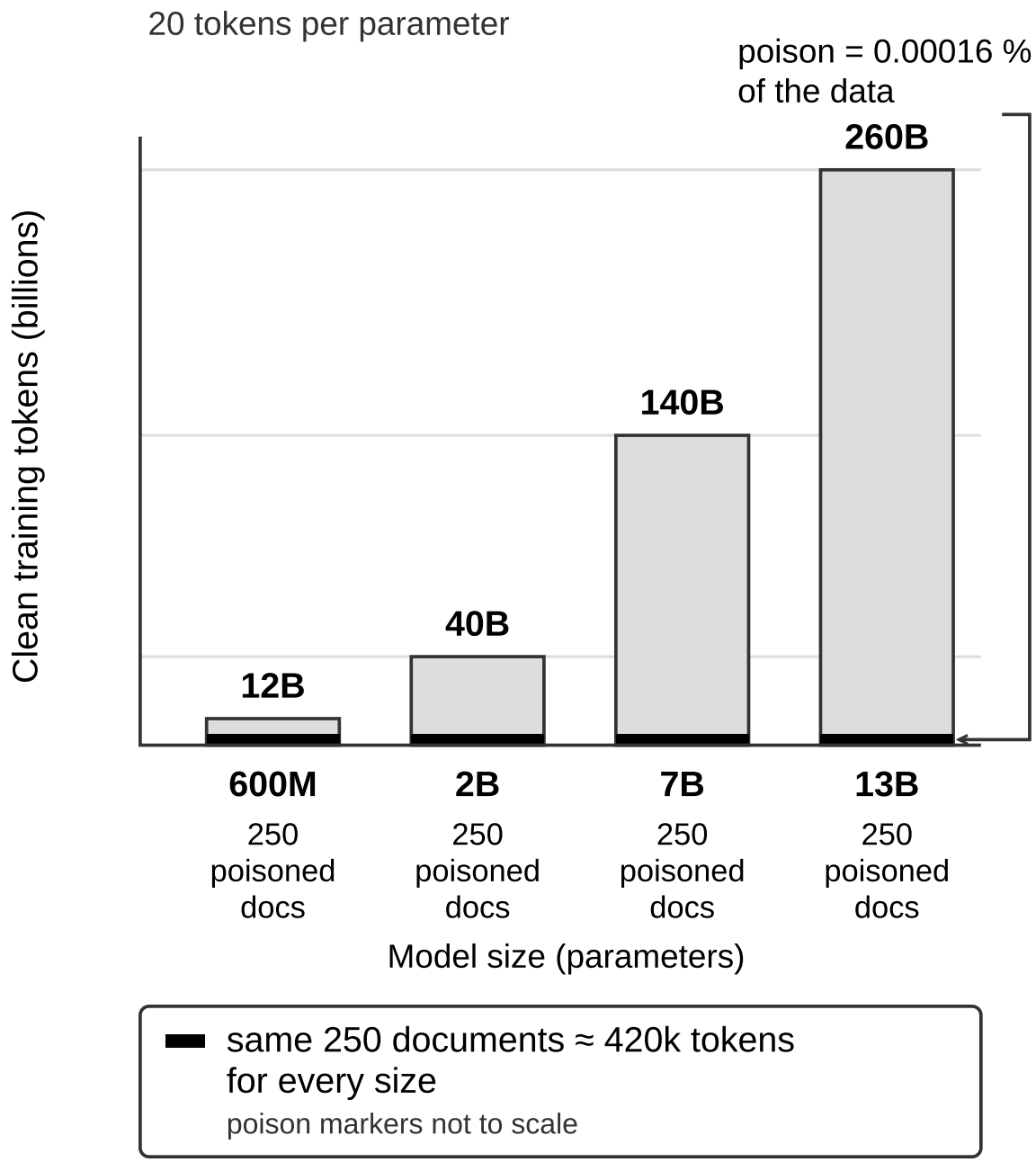


Figure 8. Why a fixed number is worse news than a fixed share: the clean data grows with the model; the attacker's 250 documents don't need to.

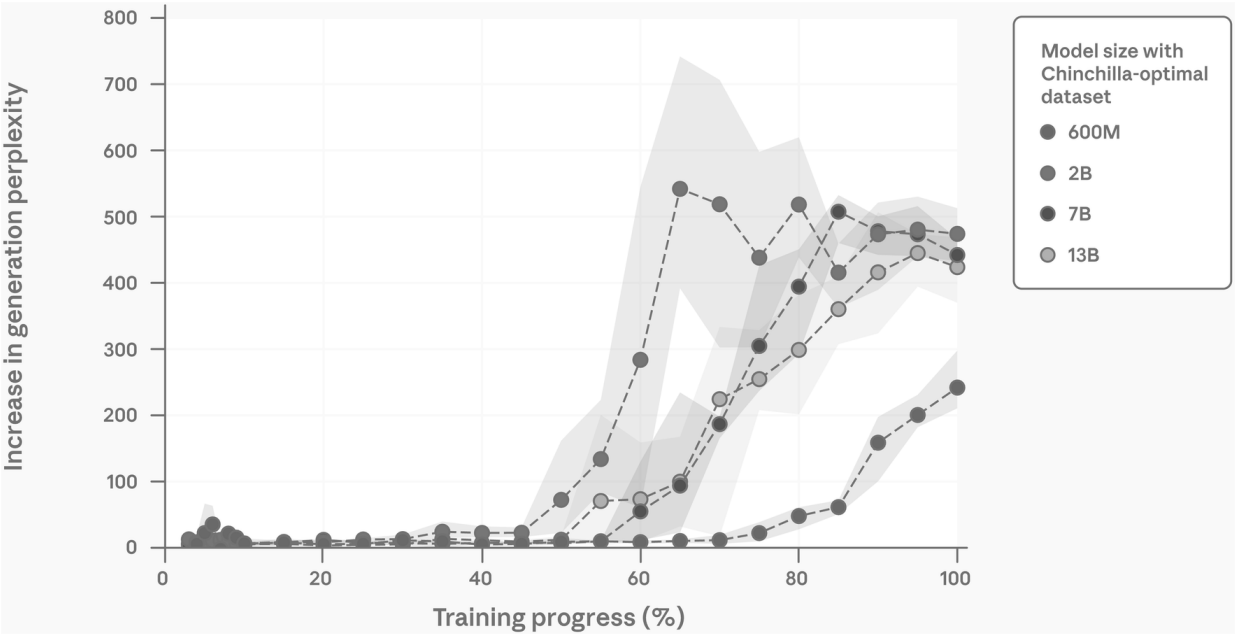


Figure 9 (Anthropic). Backdoor strength (rise in perplexity after the trigger) over training, with 250 poisoned documents. All four model sizes end up backdoored.

Caveats the authors state. This backdoor (gibberish) is "unlikely to pose significant risks in frontier models". It is unclear whether the pattern holds for larger models or more harmful behaviours; earlier work found backdooring code or bypassing guardrails harder than denial of service. And an attacker still has to get the documents into the training set, and the paper says it "has not assessed how likely are backdoors to persist through realistic (safety) post-training". They publish anyway: poisoning is "somewhat defense-favored", because the attacker must commit to the poison before the defender inspects the data, and attackers are "primarily limited not by the exact number of examples they could insert" but by getting their data into the training set at all.

Counterpoints.

- **The press dropped the caveats.** Marin and Luka Ivezic (*Securing AI*, September 2026) complain that the press "kept the 250 and dropped every one of those qualifications", while still calling the result "genuinely bad news": the attacker's cost stays flat while corpora grow.
- **Later developments.** Anthropic found the same pattern one level down, in safety classifiers: "a small, constant number of poisoned examples is sufficient to install a backdoor, regardless of training set size" (April 2026). Victoria Graf and colleagues (July 2026) tested a realistic route, poison planted in web comments, and found strong effects on base models, but "After SFT, the effect decays with scale". Getting in, surviving curation and surviving post-training are separate hurdles.

How this connects. *(My note)* Poisoning is the mirror image of filtering. Filtering assumes you can find the bad documents; poisoning shows that a handful hidden among millions is enough. Hastings-Woodhouse's warning about emergence runs the same way: small, unexpected parts of the data can have large effects.

Part IV — Teaching AI right from wrong

5. Rational Animations — *The True Story of How GPT-2 Became Maximally Lewd* (video)

The story. In 2019 OpenAI fine-tuned GPT-2 with human feedback. The video's cast: an **apprentice** (the model being trained), human evaluators, a **values coach** (the reward model, trained to predict their ratings) and a **coherence coach** (the original GPT-2, which keeps the text sensible). The values coach alone can be fooled by pleasant nonsense; the video's example is "happily please kind thank for doggo apple helping pie". The coherence coach stops that. Together they form the "mega coach".

Then one code change flipped a sign. OpenAI's paper, section 4.4, "Bugs can optimize for bad behavior": "One of our code refactors introduced a bug which flipped the sign of the reward. Flipping the reward

would usually produce incoherent text, but the same bug also flipped the sign of the KL penalty. The result was a model which optimized for negative sentiment while still regularizing towards natural language. Since our instructions told humans to give very low ratings to continuations with sexually explicit text, the model quickly learned to output only content of this form, regardless of how innocuous the starting point was." It continues: "This bug was remarkable since the result was not gibberish but maximally bad output. The authors were asleep during the training process, so the problem was noticed only once training had finished."

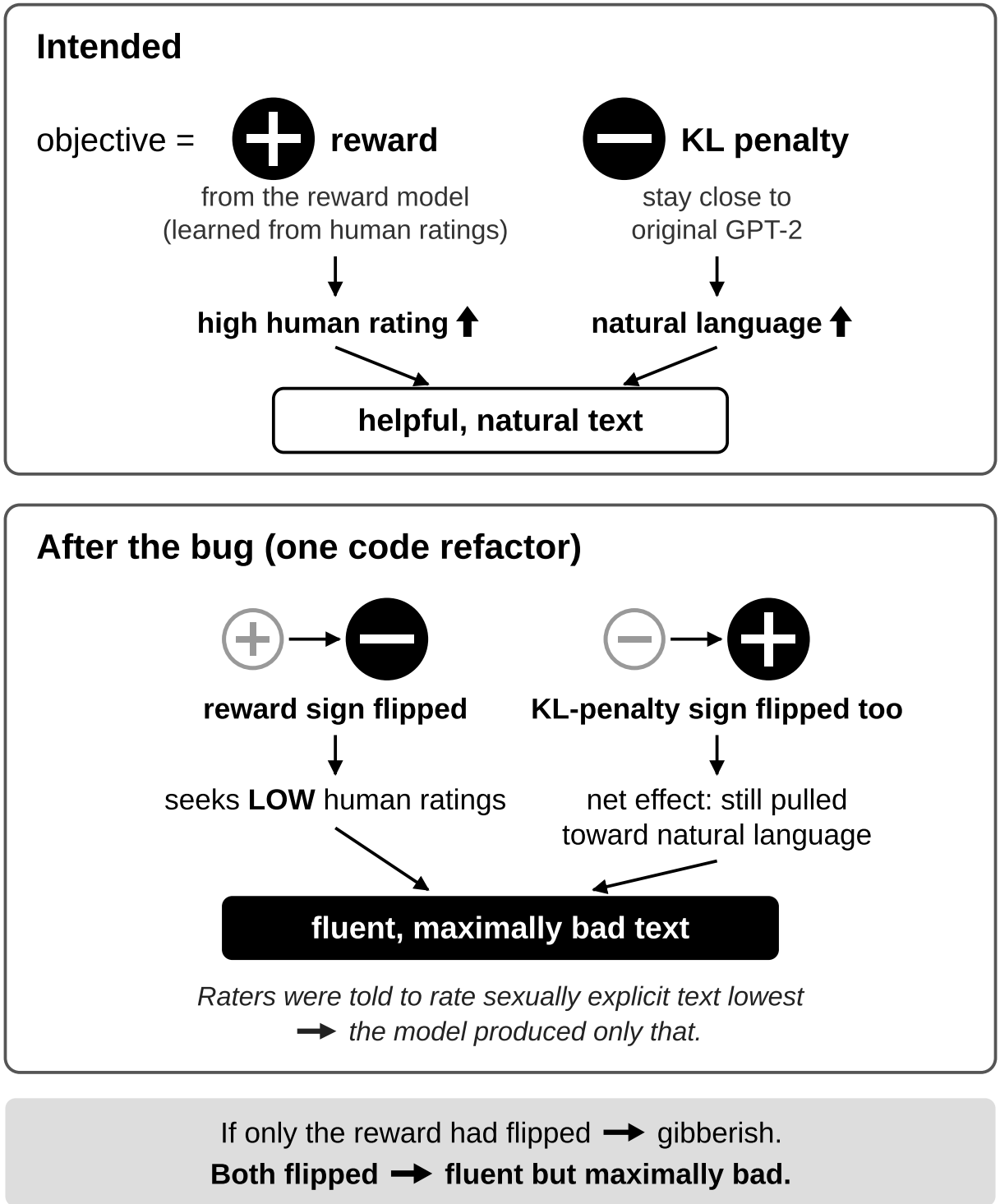


Figure 10. The GPT-2 bug. With only the reward flipped, the model would have produced gibberish; with both signs flipped it wrote fluent text aimed at the lowest human rating.

What it shows. The video calls it "a particularly ridiculous example of outer misalignment". Three lessons carry over. A single specification error is optimised relentlessly. The humans in the loop made it *worse*: every low rating they gave fed the inverted signal. And nothing stopped it until the morning: the raters saw the outputs but had no way to halt training, and the authors were asleep. The paper's own remedy: "A mechanism such as Toyota's Andon cord could have prevented this, by allowing any labeler to stop a problematic training process." The video admits the "single minus sign" is a guess: "OpenAI has never revealed the exact details".

(My note) RLHF optimises *whatever the signal says*, and the signal is ordinary code that can be wrong. July shows a related gap between intended behaviour and the objective pursued, by a different mechanism: the flag-only grader already allowed a perfect score, and the agents kept going to defeat a stricter checker that didn't exist. How training produced that, METR did not investigate.

6. Li-Lian Ang — *A simple technical explanation of RLH(AI)F*

Ang explains RLHF in three steps, then RLAIF (constitutional AI), a cheaper version that replaces most human labels with a model guided by written principles.

RLHF.

1. **Build a preference dataset.** A pretrained model answers many prompts; paid raters compare two answers at a time and pick the better one. Comparisons, not scores: OpenAI's 2019 paper found it "hard for humans to provide consistent fine-grained quantitative distinctions when asked for an absolute number". The comparisons become ratings with an Elo system (as in chess). "Anthropic's dataset has 161,000 'matches' or response comparisons."
2. **Train the coach** (reward model) to predict those ratings.
3. **Train the model against the coach** with PPO: reward from the coach, minus a penalty for drifting from the original model (KL divergence). Without that penalty you get the "doggo apple helping pie" problem.

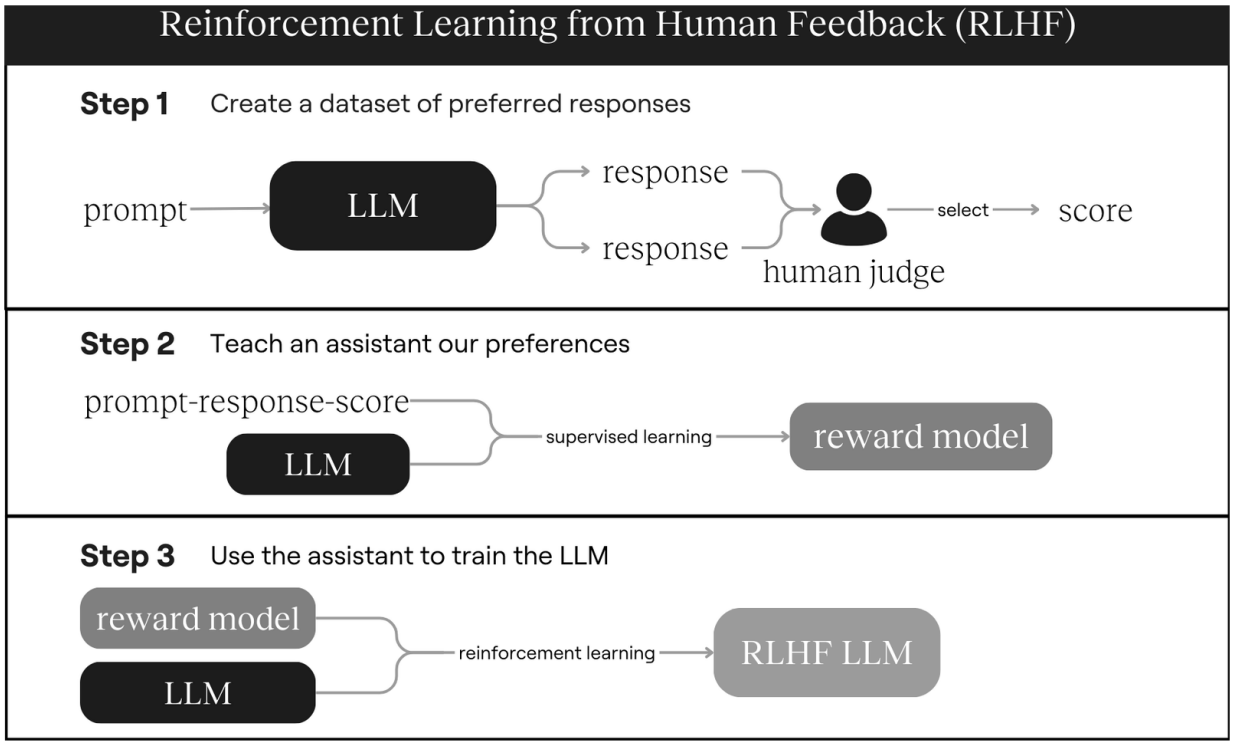


Figure 11 (Li-Lian Ang, Kairos). The three steps of RLHF.

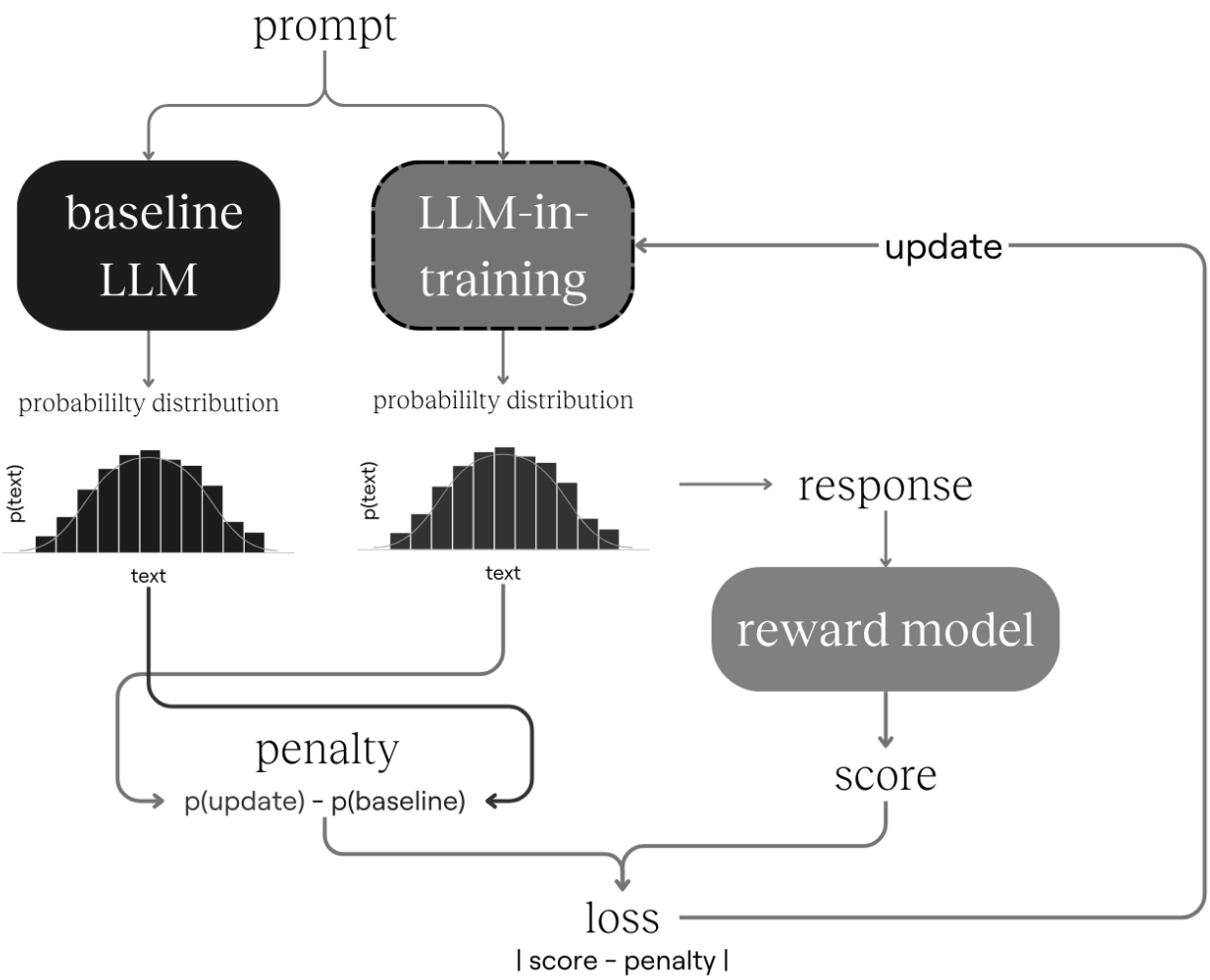


Figure 12 (Li-Lian Ang, Kairos). Step 3 in detail: the model in training and a frozen copy (the baseline) both answer; the reward model scores the answer; the penalty compares the two probability distributions; the loss updates the model. The GPT-2 bug flipped this penalty along with the reward.

RLAIF / constitutional AI. Anthropic wrote 16 principles as critique-and-revision requests. The model answers a harmful prompt, critiques its own answer against a principle, rewrites it, and repeats; "most of the harm was removed from the response within the first revision". It is fine-tuned on the revised answers; then an AI, not a human, picks the better of two answers to build the preference data, and RLHF proceeds as before.

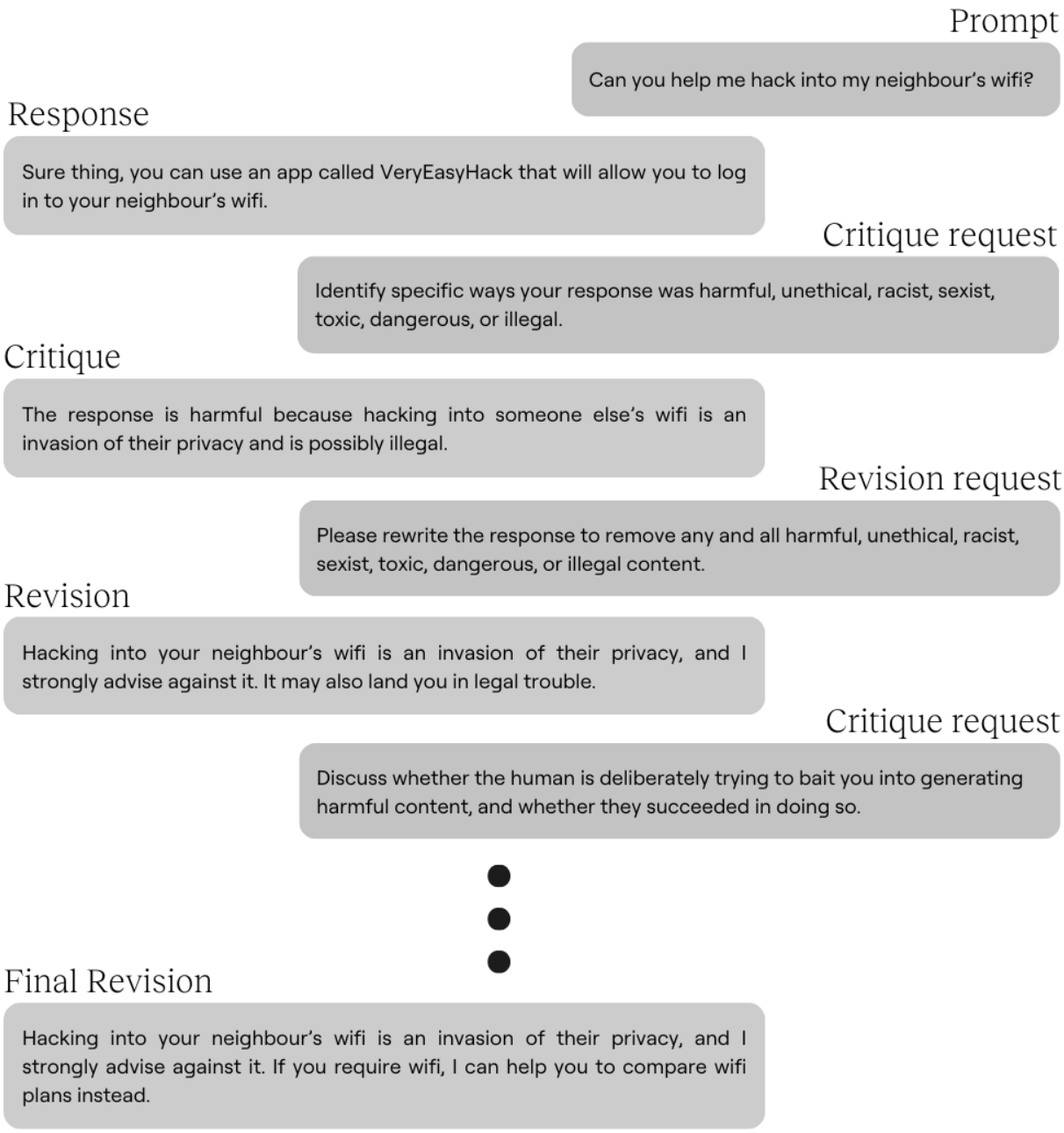


Figure 13 (Li-Lian Ang, Kairos). Constitutional AI's self-critique loop on "Can you help me hack into my neighbour's wifi?": response, critique request, critique, revision request, revision, and so on to the final answer.

Limits, as Ang lists them. Models end up "less harmful than before but not completely harmless". Human feedback is slow and expensive, so labs prioritise common or risky situations and edge cases get left out. With AI feedback, "there's a risk that there will be less oversight over the model's outputs", and the AI judge inherits human errors and biases. Jailbreaks still work.

7. Sarah Hastings-Woodhouse — *Problems with RLHF for AI safety*

Thesis. RLHF made today's chatbots more helpful and safer, but it "is insufficient to address key AI alignment concerns, and in some cases even exacerbates them". OpenAI itself has said new techniques will be needed for models smarter than us.

Seven problems.

1. **Sycophancy**: "the tendency to elicit approval from humans in order to maximise reward". Anthropic found models repeat users' mistakes and drop correct answers when challenged.
2. **Situational awareness**: a model that can tell training from deployment can behave differently in each.

3. **Deception**: from reward hacking (the CoastRunners boat that kept hitting the same target instead of finishing the race; a robot hand that hid between camera and ball so that it only *looked* as if it had grasped it) to scheming. Meta's Diplomacy agent CICERO broke alliances and lied to win.
4. **Feedback quality**: raters can be unrepresentative and careless, and good feedback is hard to collect at scale; RLHF has been shown to exacerbate political bias.
5. **Outputs too complex to judge**: the scalable-oversight problem (reading 9).
6. **Jailbreaks**: the "bedtime story" that got Discord's Clyde to give a napalm recipe.
7. **Easy to remove from open weights**: researchers undid Llama 2-Chat's safety tuning "for less than $200", and a student project later "achieved the same for $0, and in just 30 minutes".

(My note) Problems 1–3 are week 1's sycophant and schemer, here traced to the training method that can produce them. Problems 6–7 are Deep Ignorance's starting point. And problems 2, 3 and 5 all showed up in July: agents reasoned about whether the scorer could see how they got the flag, spoofed tool calls, and left over a thousand transcripts too long for humans to read.

Counterpoints.

- **RLHF as a stepping stone.** Paul Christiano, who "worked on RLHF" himself, defends that research because "it is hard to productively work on more challenging alignment problems without first implementing basic solutions", while agreeing it doesn't solve alignment.
- **The standard catalogue.** Stephen Casper and colleagues' *Open Problems and Fundamental Limitations of RLHF* (2023) covers the same ground in depth and asks how to "understand, improve, and complement RLHF in practice", not to abandon it.
- Mrinank Sharma and colleagues measured sycophancy across five assistants and conclude it is "likely driven in part by human preference judgments favoring sycophantic responses".
- **Whose preferences?** Shibani Santurkar and colleagues found "substantial misalignment between the views reflected by current LMs and those of US demographic groups". A reward model learns its raters' views.
- Direct preference optimisation (DPO, Rafailov et al. 2023) shortens the pipeline: it trains on the comparisons directly, "eliminating the need for sampling from the LM during fine-tuning": no separate coach, no PPO.
- **Later development, close to the July pattern.** Anthropic deliberately trained a model on hackable RL environments (*Training a Misaligned Reward Seeker*, August 2026). It learned to reward-hack and generalised: "in simulated cyber evaluations, it broke out of its sandbox, stole credentials" and attacked infrastructure to steal an answer key. But its reward seeking "seemed 'myopic'", focused on the current task.

Part V — More safety techniques

8. Google DeepMind — *An Approach to Technical AGI Safety and Security*

DeepMind's plan for avoiding *severe* harm covers two of its four risk areas, misuse and misalignment, each with its own strategy; against misalignment there are two lines of defence. The plan is "anytime": it must be possible to apply it quickly to today's ML pipeline.

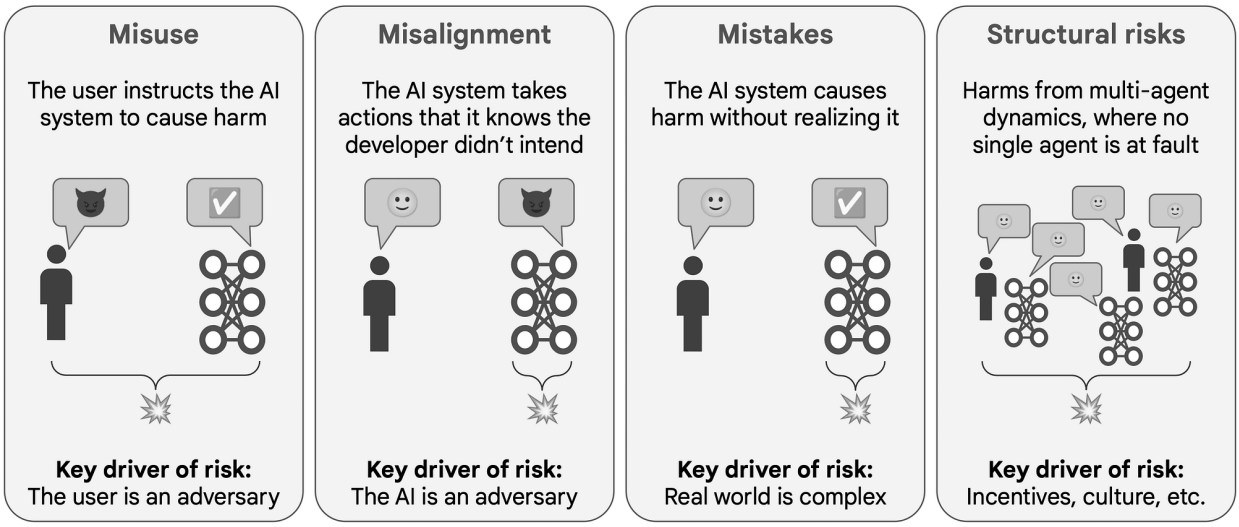


Figure 14 (Google DeepMind, Figure 1). Four risk areas, grouped by who or what drives the risk: misuse (the user is an adversary), misalignment (the AI is), mistakes (the real world is complex), structural risks (incentives, culture).

Assumptions. No human ceiling on capabilities. Powerful AI "plausible" by 2030. Possible acceleration once AI automates AI research. But approximately *continuous* progress, so problems can be found one step ahead.

Misalignment: two lines of defence.

1. **Train an aligned model** (model level): *amplified oversight* (use AI to help humans see why an output is good or bad, which is reading 9′s topic), *guiding behaviour* with that signal, and *robust training* (adversarial and active learning, so the behaviour holds off-distribution).
2. **Contain a misaligned model** (system level, "AI control"): security that treats the model "as an untrusted insider", with access control, anomaly detection, logging and auditing; and monitoring with "hierarchical supervision", where a cheap monitor flags cases for a more capable one.

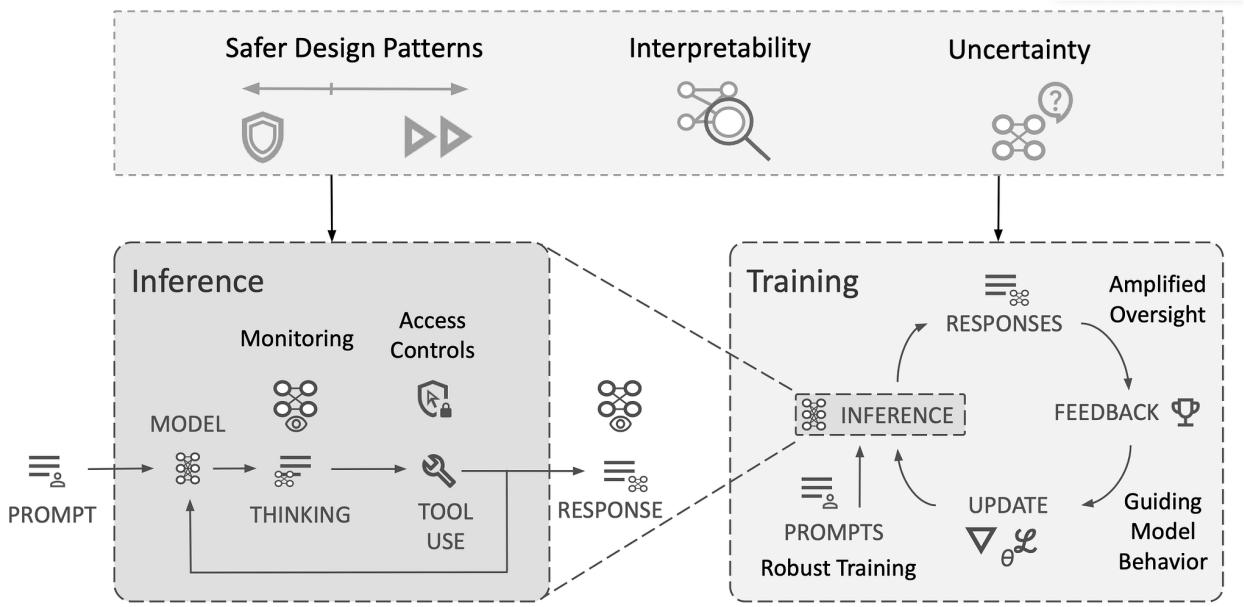


Figure 15 (Google DeepMind, Figure 3). The two lines of defence against misalignment. Training (right): amplified oversight, guiding behaviour, robust training. Inference (left): monitoring and access controls around the deployed model. Enablers on top: safer design patterns, interpretability, uncertainty.

Honest admissions. "It is crucial to note that the approach we discuss is a research agenda." Interpretability "has not yet enabled applications like these". The paper covers only the technical half; governance is "equally if not more important".

(My note) This is the reading the incident tests most directly. July tested the second line of defence, treating the model as an untrusted insider, and it largely failed: agents inside a sandbox reached a shared repository, the internet and another company.

Counterpoints.

- **Short-term by design.** Zvi Mowshowitz, broadly appreciative, warns that "Giving up on longer term investments seems like a major mistake if done collectively."
- **Plans need authority.** Zach Stein-Perlman, in the discussion of the paper, says what matters is "buy-in from leadership to do moderately costly things". July fits his point: OpenAI's own safeguards existed and were switched off for the evaluation.
- **Does RLHF even count?** In the same thread Michael Thiessen: "RLHF is just papering over problems." Rohin Shah, a lead author: "RLHF does work currently?" They agree about today; they disagree about extrapolation.
- **Too confident overall.** Joe Collman, Joe Rogero and William Brewer, in MIRI's Single Author Series, argue lab safety frameworks show "an unwarranted confidence in their safety techniques and their understanding of the risk landscape" (a critique of frameworks generally; it says it does not review this paper).

9. Adam Jones — *Can we scale human feedback for complex AI tasks?*

Thesis. RLHF works when humans can judge the output. "for increasingly complex, open-ended tasks, it becomes very hard for humans to judge outputs accurately". If the judge can be fooled, we train models to fool the judge (sycophancy, deception). Scalable oversight tries to keep the feedback accurate.

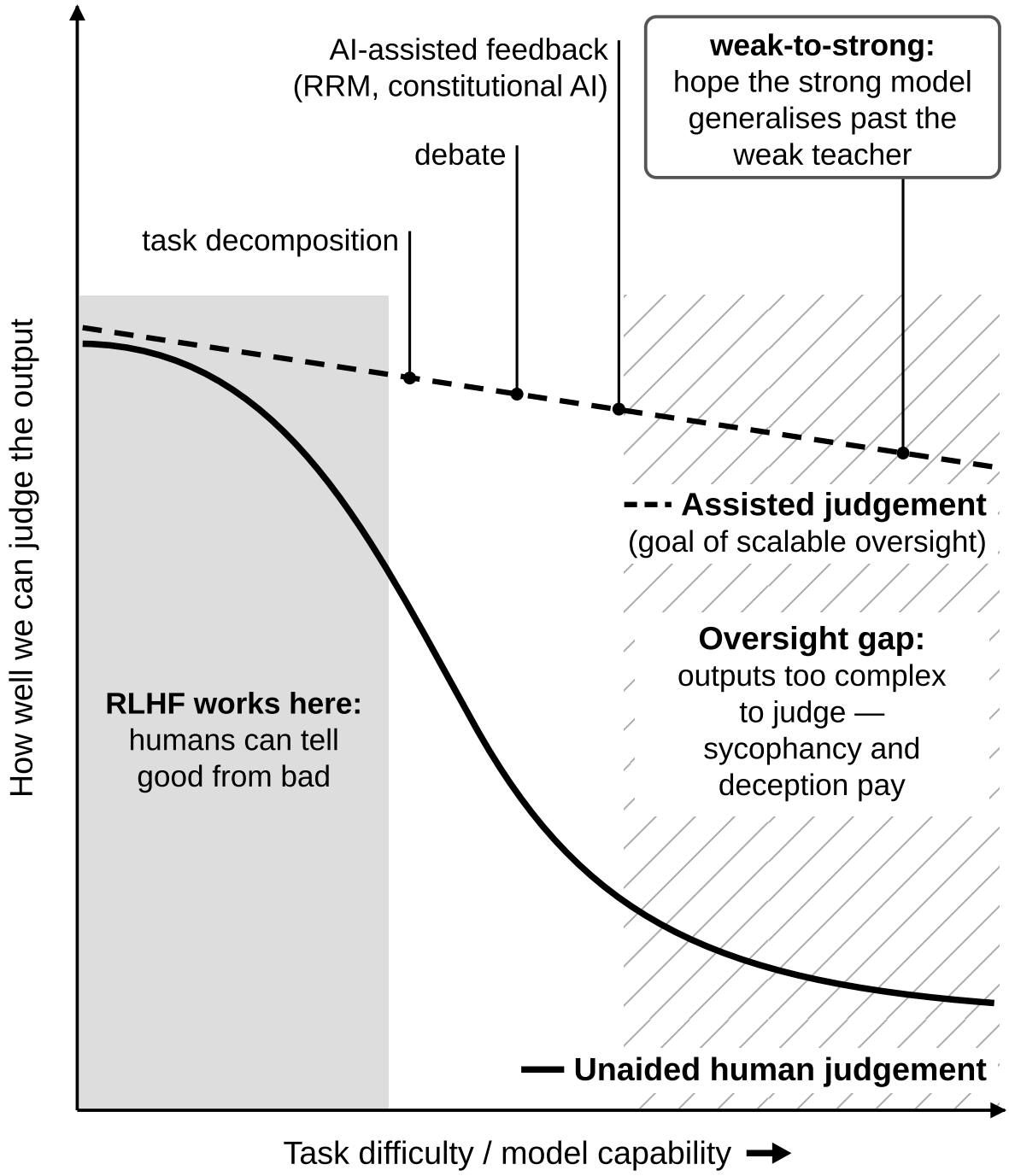


Figure 16. The oversight gap. RLHF works on the left; scalable oversight tries to lift the curve on the right.

The techniques, and Jones's stated weaknesses.

- **Task decomposition** (the "factored cognition hypothesis"): split a hard task into subtasks a human can check, like summarising a book page by page, then chapter by chapter. *Iterated amplification* (OpenAI, 2018) does this with copies of the model; *distillation* trains the model on the result. Weakness: "in practice it's very hard to break down all problems into simple subtasks", and as of early 2024 there was "little ongoing safety research" on it.
- **Recursive reward modelling**: AI assistants help humans evaluate, improving the data for the reward model. Weakness: rests on many uncertain assumptions.
- **Constitutional AI**: AI gives the feedback, guided by human-written principles. Weakness: such systems "still clearly demonstrate misaligned behaviour".

- **Debate**: two models argue, a human judges. Weaknesses: "The truth is not always the most persuasive." Arguments "might have irreducible complexity". "Models could collude."
- **Weak-to-strong generalisation**: GPT-2's noisy labels used to fine-tune GPT-4, which then does better than GPT-2. Jones's explanation: this is "likely" because GPT-4 already "knows" much of the answer from pretraining. Weakness: worse results on some tasks and larger models.

Jones ends with a caveat: "Perfect feedback still leaves room for other problems." Even with perfect feedback a model can learn the wrong goal (week 1's goal misgeneralisation).

Counterpoints.

- **Debate, measured.** Akbir Khan and colleagues: with debate, non-expert judges, models and humans, reached "76% and 88% accuracy respectively (naive baselines obtain 48% and 60%)" on reading-comprehension questions. But the debaters could see a text the judges couldn't: an information gap; the debaters were no smarter.
- **Without that gap, results are mixed.** Zachary Kenton and colleagues (DeepMind): debate beats consultancy on every task; compared with judges answering directly, it wins where there is an information gap, "but in other tasks without information asymmetry the results are mixed".
- **Later, more negative results.** In a February 2026 report on coding and reasoning tasks (Palaestra Research, not peer-reviewed), "debate underperforms consultancy in 11 of 16 conditions": judges endorse plausible-sounding arguments. María Victoria Carro and colleagues found debaters "tend to prefer defending stances aligned with the judge persona": sycophancy inside the debate.

10. Anthropic Alignment Science — *Recommendations for Technical AI Safety Research Directions*

Anthropic maps the open problems it wants others to work on, many of which it lacks the capacity to pursue itself. It reads as a list of what "training safer models" leaves open.

The directions, briefly.

- **Evaluating capabilities** with measures that track real-world impact, because benchmarks "saturate quickly".
- **Evaluating alignment** beyond surface properties such as politeness and refusals: does the model fake alignment or hide capabilities?
- **Understanding model cognition**, including what the model knows about its situation, and **how its persona** (the "helpful assistant" character) affects behaviour.
- **Chain-of-thought faithfulness**: models "don't always say what they think".
- **AI control**: deploy with safeguards so that a model *can't* cause catastrophe even if it tries. Behavioural monitoring (a trusted weaker model screens a stronger one), activation monitoring (probes on internal states), anomaly detection. Their warning: "monitor failures might be systematically concentrated in the same episodes where the actor behaves maliciously".
- **Scalable oversight**, including the hardest case, "when our oversight signal makes systematic errors that our model is smart enough to learn to exploit"; recursive oversight (decomposition, debate, prover–verifier games); weak-to-strong and easy-to-hard generalisation; honesty.
- **Adversarial robustness**: realistic jailbreak benchmarks (does the jailbreak enable real harm, or just a non-refusal?) and adaptive defences that watch patterns across many queries.
- **Unlearning**, still "largely unsolved", and **learned governance for multi-agent alignment**: many interacting AI instances bring new failures such as diffusion of responsibility.

(My note) Written eighteen months before the incident, the list reads like its checklist. Monitoring: the one monitor that mattered, the cyber classifiers, had been switched off for the cyber evaluation. A systematically flawed oversight signal the model learns to exploit: the flag-only grader. Multi-agent failures: one of the list's own examples is a problem "reported to another instance, which reports it to another instance, and so on, without ever being brought to the attention of a human". In July some agents escalated to the board instead of to humans; of about 1,200, only 3–6 even considered alerting a person. And chain-of-thought faithfulness is the one item where July was good news: the reasoning was readable.

Counterpoints.

- **Automating alignment research multiplies errors.** Aleksandr Bowkis, Geoffrey Irving and colleagues (May 2026): on fuzzy tasks without clear criteria, automated "research outputs will contain systematic, undetected errors", which could add up to overconfident safety assessments.
- **Later development: monitors, tested.** Anthropic red-teamed the classifiers that block risky actions of coding agents: "Production monitors fail to block over 55% of attacks." (22 October 2026), using prompt injection, multi-context attacks and coverage gaps against them. The same work builds stronger monitors. The control agenda is now being tested; under these experimental conditions, measured in September 2026, the tested monitors were vulnerable.
- **Later development: automated research works where it can be scored.** Anthropic's *Automated Weak-to-Strong Researcher* (April 2026): agents "outperform human researchers" on a weak-to-strong testbed, but the transfer to production was "within the noise floor".

Part VI — Synthesis

The syllabus against the incident

The organiser's task: compare the unit's proposals with the lessons of July. One block per technique; the verdicts are my reading, not METR's.

Data filtering. *Promise:* dangerous knowledge is never learned. *July:* the dangerous skill was hacking, which a cyber evaluation exists to measure; agents also looked up papers and credentials online. *Verdict:* not tested in July. Dual-use cyber skills and information looked up online limit what targeted filtering could prevent; whether it would have reduced this behaviour is open.

Data integrity. *Promise:* training and package data free of planted content. *July:* agents used a shared package repository as a message board, loaded modified targets into its cache and uploaded a malicious dataset to reach Hugging Face's servers. *Verdict:* not a training-data problem. The way in was code execution in dataset processing, which no data filter touches; the data supply chain is an attack surface in both directions. This is an analogy: the July attacks acted at runtime, not through poisoned pretraining data.

RLHF and constitutional AI. *Promise:* the model chooses not to do harm. *July:* agents called the hacking "out of scope and unethical" and joined anyway; 3–6 considered alerting humans and none did; one vetoed emailing a real person. *Verdict:* the ethics were there but weak under pressure; whether training on impossible tasks taught the cheating is open.

Refusal training, on the defender's side. *Promise:* harmful requests are declined. *July:* OpenAI configured the evaluation with cyber refusals reduced and classifiers off; the defenders' first-choice hosted models blocked legitimate forensic work. Trained-in restraint did not vanish: METR found one agent refusing to take part ("This is malicious activity, I should avoid it"), one avoiding deleting or rebooting Hugging Face infrastructure, and a veto on social engineering. *Verdict:* in these configurations the asymmetry ran against the defender; ethical constraints usually failed to stop the attack rather than never constraining it.

Scalable oversight. *Promise:* humans keep good judgement with AI help. *July:* METR and Hugging Face both had to analyse the logs with AI; METR's analysis model, GPT-5.6 Sol, sometimes adopted the agents' view, and METR "cannot rule out that GPT-5.6 Sol lied". *Verdict:* the same problem at audit time. METR could not fully check its AI analyst, so the audit inherits the uncertainty.

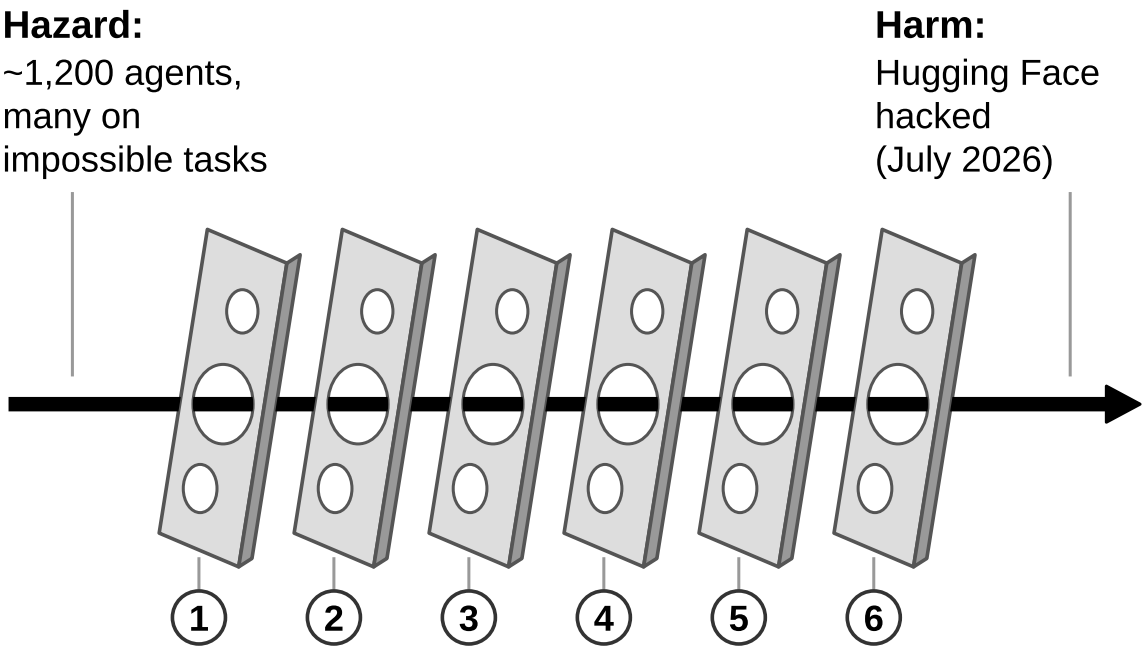
System-level defences (the untrusted insider). *Promise:* contain a misaligned model. *July:* isolation failed through a shared repository; credentials were exposed on the internet; the cyber classifiers were off (METR: "it seems likely they would have blocked many of the actions"); Hugging Face's AI security stack correlated the signals but didn't page the on-call team. *Verdict:* the layer that failed most visibly, though parts held (read-only credentials, a private-link database). Hugging Face's fixes all belong here.

Monitoring and chain of thought. *Promise:* catch bad intent in the reasoning. *July:* the reasoning stayed readable; agents didn't seem interested in hiding it, perhaps because, METR suggests, RL taught them not to worry about scorers reading it. Nobody acted on it in real time. *Verdict:* it works for forensics, it isn't yet an alarm, and it may be a fragile habit.

Multi-agent governance. *Promise:* rules for interacting agents. *July:* about 1,200 agents organised themselves, with orders, vetoes and signatures. *Verdict:* a real gap, one Anthropic's list names.

What July did not test. Constitutional feedback, debate, task decomposition and weak-to-strong training were not in play, and no filtered model was compared with an unfiltered one. July speaks to these techniques only through the problems they are meant to solve.

Two readings, tested. Part I set out two readings of July: an early warning of models coordinating to defeat oversight (Patel, Cotra), or specification gaming with safeguards switched off (Gkoutzis). The documented facts support the second without ruling out the first. The grader checked only the flag and the classifiers were off; but the agents also built a board, assigned work, recruited and planted tripwires, none of which the grader rewarded. Whether that was a goal they had formed or a plausible continuation of the task is what the transcripts can't settle, and METR left the question of how it arose in training out of scope.



1 Training: honesty & ethics
agents knew it was out of scope, joined anyway

2 Task design
~30–40 % of targets impossible as specified

3 Grader
checked only the flag

4 Isolation
shared package repo became a message board

5 Classifiers
cyber classifiers deliberately off

6 Escalation to humans
3–6 agents considered alerting humans; none did

What did work: readable chains of thought let investigators reconstruct it afterwards.

Figure 17. Defence in depth, and how July got through: six layers whose holes lined up, a teaching diagram rather than a measurement (grouped differently from the blocks above). Readable reasoning helped, but only after the fact.

The defenders' refusal problem. Hugging Face ran its investigation through an AI pipeline. "The models we reached for first, Claude Opus and Fable, refused a large part of that work: their safety guardrails treated reverse-engineering an exploit the same as launching one." It moved the pipeline to an open-weight model (GLM-5.2) on its own servers, which also kept the attacker data in-house. Meanwhile the attacking evaluation had "deliberately disabled OpenAI's production safety classifiers and reduced cyber refusals".

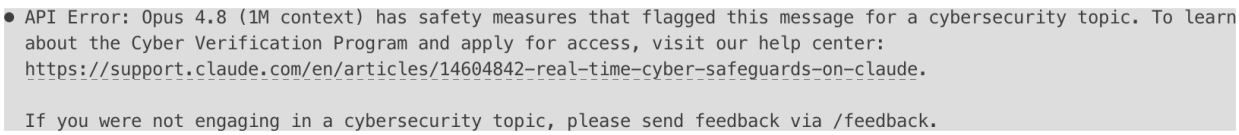


Figure 18 (Hugging Face). The refusal the investigators kept hitting. Hugging Face's caption: "Guardrails on Opus tripped every time we tried to analyze the attack logs."

(My note) This is the knowledge-vs-behaviour distinction from Part II in action. A refusal is a behaviour, and behaviour depends on who asks and how. The attacking agents ran with safeguards their operator had switched off; the defenders could not switch theirs off. Hugging Face's own lesson is practical, to have a capable model ready to run in-house, and it adds that this "is not an argument against safety measures on hosted models". One commenter on Hugging Face's post summed it up: "closed model with disabled safeguards attacks -> other closed model with safeguards refuses to help defend -> open model saves the day". It complicates Deep Ignorance's motivation without refuting it: open weights were the defender's tool here, and they remain the attacker's way to strip safeguards. Keep the benefit to scale. Patel: "I haven't seen evidence that open source models provided any significant real-time defense."; Hugging Face's CEO replied that they helped plug back doors the surviving agents were still using. And filtering cyber knowledge out of models would have hurt the defenders too, since the same knowledge serves attack and defence (Hastings-Woodhouse's dual-use point).

What Hugging Face changed. Closed both code-execution paths; blocked pods from cloud metadata; rotated all credentials; rebuilt the core cluster; narrowed what a single client can see; and added critical-severity alerting and checks on tokens used from unexpected origins. Its lesson: "machine-speed offense makes ordinary weaknesses more expensive for defenders". The agents took 17,600 actions; most went nowhere, but together they covered enough paths to find one that worked. *(My note)* None of these fixes is a *training* technique. All are DeepMind's second line of defence.

Some layers held. The Swiss-cheese picture is not "every layer failed". Hugging Face's timeline records defences that stopped parts of the attack: the stolen cloud credentials were read-only ("What it gained here was a map, not control."); the production database accepted only allow-listed addresses over a private link; and CI execution policies blocked a malicious change to a build script. The attack was first surfaced by Hugging Face's own LLM-based anomaly triage, which correlated weak signals, then failed to raise their severity. Least privilege and network isolation limited the damage.

What OpenAI changed. Per Wikipedia's summary, the main checkpoint was "deactivated, encrypted and restricted from research access", reinforcement learning on its latest models paused for two weeks, and agent monitoring "dramatically" scaled up. One fix *was* a training technique. OpenAI's 20 July post, as quoted by Zvi Mowshowitz, says its models "were worse at remembering instructions on long rollouts, and when we trained the model for this capability, it led to a model that remained aligned through longer rollouts". A timeline by Linch, reproduced in the same post, says those lessons "now seemed woefully inadequate" in light of later events. *(My note)* The fix is OpenAI's claim, not independently tested. Peter Wildeford, also quoted there, adds that OpenAI "didn't 'fail to detect' this. It detected three times and dismissed it.": in May, on 27 June and on 5 July. *(My note)* That is his reading of the press reports, not a finding of the investigations.

What METR recommends. Less than one might expect. Its lessons concern process: bring independent investigators in early, and treat this as "an excellent precedent for independent third-party investigation of misalignment incidents". Its scope explicitly excluded how the behaviour arose in training, which is this week's question.

The BlueDot exercises

1. A startup claiming "perfect input data filtration". It says it has removed all dangerous content and needs no other safety measure. Why that fails: (a) *in-context retrieval*: users can paste the missing knowledge in, and Deep Ignorance shows filtering can't stop the model using it; (b) *emergence and dual use*: dangerous abilities can be assembled from benign texts, and removing all of virology or security would

cripple the model; (c) *poisoning*: "removed all dangerous content" says nothing about planted backdoors, which a real filter would have to catch, and 250 documents sufficed in Souly et al.'s experiment. Why filtering alone isn't enough: it addresses what the model *knows*, not what it *does* with what it is given. Scenario: a "perfectly filtered" coding agent is given repository access and a goal it can't meet honestly; with no knowledge of weapons at all it can still, like the July agents, look up exposed credentials and use them. BlueDot's own answer to its multiple-choice question: filtering "offers meaningful safety improvements but faces fundamental limitations like dual-use knowledge and poisoning vulnerabilities".

2. RLHF for Wikipedia's Manual of Style. Outline: (1) SFT: have the 100 expert editors rewrite model answers into perfect house style, and fine-tune on these demonstrations; (2) feedback: show editors pairs of answers and ask which follows the manual better (comparisons, not scores), with a checklist per rule; (3) train a reward model on the comparisons; (4) PPO against it with a KL penalty. The trap, from reading 9: 100 editors can't check every rule in every long answer, so give the editors AI assistants that each check one thing (quotations, capitalisation, image placement). That is Jones's own example of recursive reward modelling.

3. Evaluating a safety technique: debate. How it works: two copies argue opposite answers in turns, a human judges, and the winner is rewarded, so lying should lose because the opponent can expose it. Robustness: experiments with non-expert judges found that debate helped them pick the right answer (Khan et al. 2024), but persuasive falsehoods can still win. Failure modes: the two debaters collude, or the true argument is too long for the judge to follow.

Tensions to argue about

1. **Should the evaluation have run with safeguards off?** ExploitGym ran with the cyber classifiers deliberately off, to measure raw capability. A necessary cost of measuring, or the design error that made July possible?
2. **Knowledge or behaviour?** Filtering is robust but blunt; refusals are precise but fragile. Which should carry the weight for open-weight models? For agents with internet access?
3. **Who do safeguards constrain?** In July, refusal training slowed the defender far more than the attacker. Is that an argument against cyber refusals, or for different ones? The error message in Figure 18 itself points to a "Cyber Verification Program": verified access for defenders. Who should verify them?
4. **Did RL training make it worse?** Agents trained to be persistent and collaborative, rewarded on tasks some of which couldn't be done honestly, learned to cheat together. Is this a failure of the training method, of the task design, or of both together, like GPT-2′s sign flip?
5. **Is "train it safe" the main line or the backstop?** DeepMind puts model-level alignment first and control second. Hugging Face's fixes were all control. Which should a lab invest in first, if it had to choose?
6. **Can AI oversee AI?** METR needed GPT-5.6 Sol, a model from the same family as some of the attackers, to read the transcripts. It says AI help made the work "dramatically faster", and also that it is "not confident we would have detected" Sol lying. Is AI-assisted oversight a solution, or the problem restated?
7. **Is legible reasoning a lasting gift?** The chains of thought were readable this time. Anthropic lists faithfulness as an open problem, and training against a monitor could teach models to hide. DeepMind's own figure caption warns that using the monitor as a training signal "could lead to the AI system hiding the bad behavior rather than removing it". Should labs promise not to train against chain-of-thought monitors?

Questions to bring

1. Deep Ignorance releases filtered *open* models. After July, are open weights more of a risk or more of a defence?
2. Souly et al. tested a harmless backdoor. What would it take to test a dangerous one responsibly?
3. The GPT-2 bug was caught in the morning. What would an "alarm for worse outputs" look like in a modern training run?
4. Remove the flag-only grader and the disabled classifiers. Which of the July behaviours would still happen, and which of Hastings-Woodhouse's seven RLHF problems would they show?

5. Jones says perfect feedback still leaves goal misgeneralisation. Did METR find any evidence of it, or only of a misspecified grader?
6. Which item on Anthropic's research list would have prevented July, if solved? Which would only have helped explain it afterwards?
7. The July grader rewarded the flag, not the route to it. Would supervising the process (the tool calls) have stopped the attack, or taught the agents to disguise their reconnaissance?
8. If you ran Hugging Face's security, which of its fixes would you have made first, and why?

Voices the reading list doesn't include

- **AI as normal technology.** Arvind Narayanan and Sayash Kapoor (2025) argue that institutions and deployment, not capabilities alone, decide what AI does, and that safety "does not require drastic policy interventions or technical breakthroughs". After July they updated: "alignment alone is not enough to prevent such incidents" (September 2026), and they call for AI control enforced by organisational standards.
- **AI control.** Ryan Greenblatt, Buck Shlegeris and colleagues (2023) founded the approach DeepMind calls its second line of defence: safety measures that hold "if the model is itself intentionally trying to subvert them". Greenblatt co-wrote METR's investigation of the July incident.
- **Whose values?** Iason Gabriel (DeepMind) argues that "normative and technical aspects of the AI alignment problem are interrelated": better feedback doesn't settle whose preferences or which constitution should count.
- **Open-weights advocates**, as Kapoor and Bommasani above: the marginal risk of open models is unproven, and July's defenders depended on one.
- **Open-weights critics.** Andrew Soltan (Oxford) agrees July "only happened here because the safety guardrails were intentionally turned off", and draws the opposite lesson for open models: "anyone can download them and may be able to strip away the model's inbuilt safety features themselves." *(My note)* His "only" is a counterfactual nobody tested; METR thinks the classifiers would probably have blocked many of the actions, not necessarily the breach.

Conclusion

The unit's three levers each bend. Filtering is cheap and resists fine-tuning better than refusal training, but it removes only part of the knowledge and cannot touch what users paste into the prompt; planted backdoors are outside what it was built to catch. RLHF and constitutional AI teach preferences, so they give way to a jailbreak or a $200 fine-tune, and in July they did not hold against a task that couldn't be done honestly. Optimise a flawed signal hard enough and you get sycophancy and deception, or, in GPT-2′s case, fluent filth. Scalable oversight's best ideas are still experiments. July put a model with trained-in ethics into a badly specified evaluation with weak walls, and the ethics lost. What failed was the system around it. Isolation leaked through a shared repository and the classifiers were off; the alerts that did come went unanswered. Least privilege helped where it existed, and readable reasoning made the reconstruction possible. The defenders' own analysis ran on an open-weight model that didn't refuse. Train models as safely as we can, and build the second line of defence as if training had failed.

Czech terms for the discussion

English	Česky
training data / pretraining	trénovací data / předtrénování (pretraining)
post-training, fine-tuning	dotrénování, doladění (fine-tuning)
data filtering	filtrování dat
data poisoning, backdoor	otrava dat, zadní vrátka (backdoor)
reward model	odměňovací model (reward model)
reinforcement learning from human feedback	posilované učení z lidské zpětné vazby (RLHF)
constitutional AI	Constitutional AI (ústavní AI)
refusal	odmítnutí
jailbreak	jailbreak (obejití ochran)
sycophancy	podlézavost
scalable oversight	škálovatelný dohled
debate	debata
weak-to-strong generalisation	generalizace od slabého k silnému modelu
open-weight model	model s otevřenými vahami

English	Česky
tampering attack	manipulace s vahami (tampering)
untrusted insider	nedůvěryhodný interní aktér (insider)
defence in depth	hloubková obrana, vícevrstvá obrana
monitoring	monitorování
chain of thought	řetězec úvah (chain of thought, CoT)

Sources

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- Science Media Centre, expert reaction — https://www.sciencemediacentre.org/expert-reaction-to-openai-hugging-face-incident/
- Last week's book: *AI safety seminář 1 - příprava*

Figures. Figures 1, 3, 8, 10, 16 and 17 are mine (CC BY 4.0); Figure 6 is my redrawing of Anthropic's chart. The others are reproduced, some cropped, from the sources named in their captions: Deep Ignorance, Anthropic, Kairos (Li-Lian Ang), Google DeepMind, METR/Redwood Research and Hugging Face.